Assurance from Junior Allies: Cooperative Management of Adversaries Below the War Threshold

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Abstract

Conventional wisdom on asymmetric alliances focuses on deterring war, thus centers the senior ally's capabilities, casting junior allies as dependent. This paper examines cases where all-out war is effectively deterred but adversaries still engage in low-intensity provocations. I argue that here, assurance rather than coercion shapes adversary behavior, and junior allies can independently play a role. An original monthly dataset of U.S. and South Korean policy toward North Korea from 1991 to 2011, coded across five dimensions (economic aid, diplomacy, sanctions, rhetoric, military posture), shows that at the aggregate level only South Korea's orientation is marginally significant, while U.S. policy and inter-ally alignment are not. Disaggregating, South Korea's cooperative rhetoric and U.S. diplomatic engagement are independently associated with fewer provocations. These results suggest that managing adversaries below the war threshold depends not only on patrons' threats but on how each ally deploys specific assurance instruments.

Most recent version: https://kohyunsup.github.io/jmp.pdf

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1 Introduction

When the United States held its North Korea policy largely steady across the Biden administration, North Korean provocations nonetheless surged from roughly 1.3 to 4.3 per month (CSIS Beyond Parallel 2025), in tandem with South Korea's transition from Moon Jae-in to Yoon Suk-yeol. Under Moon, Seoul pursued a cooperative approach toward Pyongyang, emphasizing engagement and de-escalation; under Yoon, it shifted to a far more coercive stance, even labeling North Korea an enemy in its first Defense White Paper (Ministry of National Defense 2023). The puzzle is that the patron's posture barely moved, yet the adversary's behavior did, tracking the junior ally instead. That points to a role for junior allies that existing frameworks rarely examine: active managers of adversary behavior through their own policy choices. This recent pattern echoes a broader dynamic that this paper analyzes systematically using the U.S.–South Korea alliance from 1991 to 2011.

Alliance and extended deterrence scholarship offers limited guidance here, because it largely treats alliances as unitary actors or centers the senior ally's capabilities and resolve, casting the junior ally as a recipient of protection rather than an active player vis-à-vis the adversary (Waltz 2010; Snyder 2007). This emphasis is well founded on the threat side, where deterrence rests on aggregate capability and the junior ally is outmatched. But threat alone does not make deterrence work. Assurance, its complement, turns on whether the adversary expects restraint to be reciprocated (Schelling 1966; Huth 1988), and on that margin the junior ally is not disadvantaged, and may be better placed than the distant patron. The distinction matters most below the war threshold. Deterring war is largely a matter of threat, where the patron dominates; but once war is off the table, what remains is the day-to-day management of provocations, and this sub-war space, not major war, is where the majority of adversary interactions actually occur.

This paper examines that space. I focus on asymmetric alliances in which escalation to all-out war is already off the table, yet adversaries still engage in recurring hostile actions short of open war. Here the central question is no longer who prevents major war, but how

each ally shapes the adversary's daily decision to provoke. Building on the classic insight that effective deterrence pairs firm threats with flexible assurances (Huth 1988), I argue that the two zones have different logics. War-deterrence runs on threat and aggregate capability, where the patron naturally dominates. Sub-war management runs on assurance, and in that zone the junior ally, structurally advantaged by proximity, higher issue salience, and frequent interaction with the adversary, can play a meaningful, independent role that patron-centric frameworks would not lead us to expect. This does not mean the junior ally replaces the patron below the threshold; it means the junior ally is no longer sidelined there.

The reason is that in this setting additional coercion faces diminishing returns, a condition I call coercion saturation. By coercion saturation, I mean a condition in which the adversary already expects catastrophic punishment for major escalation, so additional coercive moves change its expectations about ongoing pressure more than its expectations about the costs of war. Once that point is reached, more coercive policies mostly reiterate a threat the adversary already understands, and a posture dominated by coercion can even sour its expectations, signaling that pressure will tighten regardless of restraint (Pauly 2024). Cooperative tools operate on a different margin, communicating that restraint will be acknowledged and potentially rewarded. In other words, under coercion saturation, coercive policies degrade the adversary's expectations of restraint and thus fuel sub-war provocations, whereas cooperative policies improve those expectations and dampen them, and the junior ally is the member well-positioned to supply such assurances.

Based on this logic, I advance two claims. First, in asymmetric alliances where all-out war is structurally deterred, more cooperative policy toward the adversary is associated with fewer sub-war provocations, whereas more coercive policy is associated with more provocations. Second, junior-ally policy can independently affect the adversary's behavior.

I test these claims using the U.S.–South Korea alliance and North Korean provocations from 1991 to 2011, a period when all-out war on the peninsula was consistently deterred but sub-war provocations were frequent. I introduce a portable, transparent coding framework

and use it to construct a new monthly dataset that codes U.S. and South Korean policy orientations toward North Korea independently across five generic instruments: economic aid, diplomacy, rhetoric, sanctions, and military posture. This granularity captures behavioral variation that aggregate annual measures or binary hawks-versus-doves classifications would miss, and by coding each ally separately it allows a direct test of whether cooperative policies are associated with fewer provocations and whether the junior ally's effect is independent of the senior ally's.

The results are consistent with both claims. At the aggregate level, only South Korea's overall policy orientation shows marginal significance, while U.S. policy and inter-ally alignment are not statistically significant. Disaggregating by dimension reveals that South Korea's cooperative rhetoric and U.S. diplomatic engagement are the only two channels that consistently reduce the likelihood of North Korean provocations, and South Korea's rhetorical stance retains its significance even after controlling for the full set of U.S. policy instruments, indicating that the junior ally contributes independently rather than merely transmitting the senior ally's influence. These findings complement the assurance tradition's insight that threats require credible assurances, extending it with a disaggregated picture of how such assurances operate below the war threshold, while challenging the patron-centric emphasis on capabilities and resolve: below the war threshold, cooperative signaling by a junior ally plays a distinct and measurable role in shaping the adversary's decision to provoke.

This study advances the empirical study of alliances and deterrence, analyzing how each ally's specific policy instruments independently shape adversary provocations below the war threshold, and building on scholarship that has examined the role of assurances alongside threats, junior-ally agency within alliances, and approaches to measuring alliance contributions (Davis 2000; von Hlatky 2013; Kydd and McManus 2017; Bernhardt and Sukin 2021; Blankenship 2021; Pauly 2024). It makes three further contributions. First, it introduces an original monthly dataset that disaggregates each ally's policy across five instruments; because the coding scheme relies on generic policy categories rather than case-specific knowledge, it

is adaptable to other similar settings such as U.S.Taiwan-China. Second, it develops a theory of coercion saturation that explains why cooperative tools reduce provocations while coercive tools face diminishing returns under a fixed strategic deterrent backdrop. Third, it provides evidence that a junior ally can independently shape adversary behavior through its own cooperative policies. These findings also carry a policy implication: rather than pressing junior allies to match the patron's coercive capabilities in burden-sharing debates, recognizing that junior allies can use the everyday assurance tools they are structurally suited to deploy may improve deterrence below the war threshold.

2 Assurance, Junior Allies, and the Sub-War Gap

Alliance politics builds on balance-of-power and deterrence theory, which conceptualize alliances as devices for aggregating military capabilities to confront external threats and prevent war (Waltz 2010; Walt 1985; Mearsheimer 2001; Brooks and Wohlforth 2005). Conventional approaches therefore emphasize coercive tools such as military capabilities (Huth and Russett 1984; Huth 1988; Leeds 2003; Signorino and Tarar 2006) and resolve, understood as the willingness to bear costs in defense of allies (Fearon 1995; Fuhrmann and Sechser 2014; McManus 2017), and canonical definitions of deterrence as "the threat or use of force to dissuade an adversary from taking an undesirable action" reinforce this focus on coercion and war-prevention outcomes. Yet classic deterrence theorists have long noted that coercion is not purely about punishment: Schelling highlights that effective coercion combines threats with corresponding assurances that an adversary can avoid pain by complying (Schelling 1966), while Huth's work on extended deterrence shows that a "firm-but-flexible" policy can strengthen deterrence by signaling both determination and restraint (Huth 1988). Subsequent scholarship has developed this insight by treating assurances and promises as integral to influence rather than peripheral to threats (Davis 2000), and by modeling threats and assurances as complementary tools in crisis bargaining (Kydd and McManus 2017). Re-

cent contributions on the assurance dilemma further underscore that deterrence performance depends on how states combine threats with credible assurances, and that mismanaging assurances can backfire even when coercive capability is ample (Pauly 2024). For the most part, though, these insights have been worked out where the stakes are highest—discrete crises with a single defender deciding between war and peace—leaving the everyday management of provocations below that threshold less examined.

This confinement reflects a deeper orientation in the field. Most alliance and extended deterrence research evaluates deterrence in terms of war-versus-peace outcomes and immediate deterrence success. Although rationalist accounts imply a wide bargaining range between full cooperation and all-out war, in which conflict short of war can unfold before bargaining breaks down (Fearon 1995; Reiter 2003), empirical work has overwhelmingly concentrated on whether alliances prevent war rather than on how they manage recurring low-level provocations short of war. This emphasis is closely connected to the field's focus on external balancing and aggregate capability: because alliances are understood as mechanisms for pooling power, it is understandable that scholars have centered the senior ally's military capabilities and deterrent posture. Extended deterrence by definition highlights a powerful patron that "extends" protection to a weaker protégé (Huth and Russett 1984), and many studies therefore model outcomes as functions of aggregate power mostly provided by senior allies, while junior allies appear mainly as recipients of security worried about abandonment and entrapment (Cha 2000; Snyder 2007; Kudo 2025), as potential free-riders in burden-sharing debates (Sandler and Hartley 1999; Blankenship 2020), or as sources of moral hazard under the alliance umbrella (Benson et al. 2014). A growing literature on small-state agency and asymmetric alliances has challenged the view that junior allies are purely passive, showing that they work to manage autonomy (von Hlatky 2013; Womack 2015), influence patrons (Long 2017), and shape alliance decisions (Wong 2020) despite capability asymmetries. But this work is focused on the intra-alliance relationship rather than on the adversary; it follows naturally from the war-prevention and patron-focused concerns animating the field, yet it

leaves open whether and how junior allies independently affect adversary behavior below the war threshold.

Measurement practices reflect the same emphases. Quantitative work on alliances and deterrence typically relies on input measures such as defense spending, spending as a share of GDP, or troop numbers as proxies for contribution to collective security and deterrence (Sandler and Hartley 1999; Blankenship 2021). These aggregate metrics are central to both deterrence and burden-sharing debates, but they are at best noisy proxies for actual deterrent effects (Carroll and Kenkel 2019; Rubin 2023); spending targets like the "2 percent of GDP" benchmark say little about the strategic value of what allies actually do (Kalnins 2017; Alozious 2022). This pattern is understandable: the field has measured deterrence through aggregate inputs largely because it focused on war prevention and patron capabilities, which made input measures seem sufficient. One consequence is that we still lack measures capturing how each ally uses specific policy instruments toward adversaries, making it difficult to see how those tools shape adversary behavior below the war threshold.

The field thus has powerful accounts of how alliances prevent major war but far less on the everyday management of sub-war provocations, where most adversary interaction actually occurs; and because prevailing measures privilege aggregate capability, junior allies' contributions there remain hard to observe. The next section develops a theory of how alliance members, and the junior ally in particular, shape adversary provocations once war is already structurally deterred.

3 Theory and Hypotheses

3.1 Scope condition: war prevention and sub-war conflict

As discussed above, existing studies on deterrence have made significant progress in analyzing how alliances help prevent war. The primary focus of alliance and extended deterrence research has been on whether and how states avoid all-out war. Yet the world does not

operate on a simple war–peace dichotomy; between war and peace lie multiple stages of conflict in which ongoing risk management takes place, and in which states mix coercive threats with assurances in managing adversaries below the war threshold. Most alliance-adversary relationships, moreover, spend the overwhelming majority of their duration in this sub-war zone rather than in open conflict, so it is here that the bulk of adversary interaction actually occurs.

This study focuses on asymmetric alliances in which escalation to all-out war is currently off the table, yet adversaries still pursue limited aggression below the threshold of open conflict. In many contemporary cases, a central structural constraint is the alliance's strategic deterrent posture: the senior ally's nuclear forces and conventional superiority make full-scale war prohibitively costly for the adversary, and the alliance's commitments raise the expected costs of a direct attack on the junior ally to extreme levels. The aim here is not to argue that extended deterrence is always effective or that allies are fully reassured; rather, the analysis takes as given those periods in which major war has not occurred and conflict is concentrated in recurring, limited confrontations below the war threshold. This displacement of conflict from the strategic level to the sub-war level reflects the logic of the stability-instability paradox, in which stability in the strategic balance makes lower-level provocation more likely (Snyder 1965; Jervis 1989). In such settings, adversaries redirect their efforts into sub-war provocations—missile and weapons tests, border incidents, infiltrations, and incursions—that probe alliance red lines and impose costs without crossing the threshold of open conflict.

Because war typically results from the escalation of tensions rather than emerging out of nowhere, managing the sub-war zone is not a separate concern from war prevention but part of it: wars grow out of accumulated, mismanaged provocations, so reducing the frequency and intensity of those provocations is itself a contribution to keeping war off the table. This study therefore takes the deterred threshold as given and asks what lowers provocations within the sub-war zone, rather than modeling the point at which provocations would tip

into war. Where exactly that tipping point lies is an important question, but it is not the one pursued here; the analysis holds the deterred condition fixed and examines how alliance members can influence the adversary's decision to engage in, or refrain from, provocations by combining coercive threats with assurances about the consequences of restraint.

In addition, this study assumes that the adversary monitors and responds to allied policy choices. While internal dynamics undoubtedly influence adversary behavior, the logic of deterrence presupposes that the adversary is at least partly responsive to external conditions; otherwise, there would be nothing to threaten or assure. Empirical work on North Korean propaganda supports this assumption, showing that Pyongyang's threatening rhetoric responds systematically to changes in its security environment and that surges in such rhetoric precede military provocations (Sukin 2022).

3.2 Coercion saturation and assurances

Largely because of the war-prevention focus identified above, most deterrence scholarship emphasizes threats of punishment and other coercive tools, with assurances or accommodation treated as secondary. At first glance, the concern that cooperative gestures may be read as weakness—and thereby invite more aggression—seems reasonable, especially in war-prevention scenarios where perceived weakness could lower the expected costs of attack. In settings where escalation to major war is still plausible, strengthening threats may indeed be the primary way to deter.

This study, however, focuses on settings in which major war is already effectively off the table, commonly because of the alliance's strategic deterrent posture described above. As set out in the scope condition, this is the environment the stability-instability paradox describes: sub-war provocations are not a residual anomaly but the predicted consequence of a stable strategic balance.[1]

[1]The classic paradox is dyadic and assumes mutual second-strike capability. Here I invoke its central logic—strategic stability displacing conflict downward—rather than its original symmetric setup; recent work extends the paradox to asymmetric and contemporary East Asian settings (Hiim and Tunsjø 2025;

In these environments, existing nuclear and conventional forces provide an extreme form of coercive threat, and additional coercive measures face diminishing returns. This follows the general logic of diminishing marginal returns: once a coercive posture already communicates a credible threat of catastrophic punishment, each further increment of coercion adds progressively less deterrent value. The point has a psychological counterpart in prospect theory's principle of diminishing sensitivity, whereby the subjective impact of a given change shrinks as one moves further from the reference point; far out in the domain of losses, an adversary already facing catastrophic threatened costs registers little additional pressure from added coercion (Kahneman and Tversky 1979; Tversky and Kahneman 1992). More troops, larger exercises, or additional sanctions mostly reiterate what the adversary already knows: that it is threatened. In this sense, the coercive dimension of the relationship has reached a ceiling, a condition I refer to as *coercion saturation*.

Under coercion saturation, the central uncertainty for the adversary is not whether crossing the war threshold would be costly, but what happens if it shows restraint. Does the alliance continue to increase pressure regardless of the adversary's behavior, or is there a path in which restraint is acknowledged and met with some degree of accommodation on the alliance side? This is fundamentally an information and expectation problem. It reflects a long-standing insight in coercion theory: effective coercion combines threats with corresponding assurances that compliance will spare the adversary the threatened harm, so that a threat lacking a credible assurance fails to change behavior (Schelling 1966). Recent work conceptualizes this as an assurance dilemma, showing that coercion often fails not because threats are insufficient but because targets doubt that compliance will be rewarded with restraint (Pauly 2024).

This points to a distinction central to the present argument: between assurance and mere accommodation. Assurance is conditional. It is the credible promise that if the adversary shows restraint, punishment will be withheld; it is the complement of a threat, not a substi-

Roehrig 2016).

tute for it. Accommodation, by contrast, is an unconditional benefit conferred regardless of the adversary's behavior, and it is the kind of concession that the conventional worry treats as weakness inviting further aggression. What determines whether a cooperative move is read as assurance or as accommodation is the deterrent backdrop. When the threat of catastrophic punishment is credible and fixed, a cooperative move is legible as the conditional half of a threat-plus-assurance pair: the punishment remains certain, and the cooperation is what restraint earns. Absent that backdrop, the same move is merely accommodation, and may indeed invite more aggression. Coercion saturation is therefore not only the condition under which added coercion stops working; it is also the condition that converts cooperative policy into credible assurance, because the saturated deterrent ceiling supplies exactly the standing, believed threat that assurance requires.

It follows that when cooperative signals are absent and coercive moves dominate, the adversary has little reason to expect that restraint will be rewarded; this worsens its expectations and can encourage more sub-war demonstrations of resolve rather than fewer. Cooperative policy choices, such as conciliatory gestures or limited accommodation, can instead function as assurances: they clarify that, while war-level threats remain in place, the alliance is prepared to moderate day-to-day pressure or offer limited cooperation if the adversary refrains from provocations, rather than tightening the screws automatically.

In this framework, threats and assurances are complementary rather than competing. Threats at the war level establish the high costs of escalation, but once those threats are fully credible, the marginal deterrent value of additional coercion is limited; beyond that point, extra coercive moves primarily worsen the adversary's expectations about restraint and can encourage more sub-war provocations. Variation in below-the-threshold provocations should instead depend more on whether the alliance adopts more coercive or more cooperative policy toward the adversary.

Not all cooperative policy carries assuring content in equal measure, however. Given a credible deterrent backdrop, a cooperative move reduces provocations most reliably when

three further conditions hold. First, the adversary must be able to observe the signal: it must be public and legible enough to register as a deliberate policy choice rather than noise. Second, the adversary must regard the sender as able to follow through, so that the signal reflects real discretion over the adversary's environment; the deterrent backdrop makes the threatened punishment real, and this condition concerns whether the sender can credibly deliver the reciprocal restraint. Third, the adversary must be able to see how its own restraint would be reciprocated, so that the signal maps restraint onto a concrete reduction in pressure or a specific accommodation. Cooperative measures that fail these conditions may be conciliatory in direction yet carry little assuring content. This distinction implies that assurance is a property of particular policy signals under a credible deterrent backdrop, not of cooperative gestures in general, and that instruments will vary in how reliably they carry it.

> ***H1 (Assurance and cooperative policy).*** *In asymmetric alliances where all-out war is structurally deterred, more cooperative policy toward the adversary—functioning as assurance under the credible deterrent backdrop—is associated with fewer sub-war provocations, whereas more coercive policy is associated with more provocations.*

3.3 The junior ally's independent role

If all-out war is already off the table, why can the alliance not simply rest on that achievement? The answer is that a war-free status quo is not a risk-free one. Even when large-scale conflict is kept off the table, primarily by the senior ally's nuclear umbrella and conventional superiority, the adversary continues to probe through border incidents, incursions, and other provocations that fall short of triggering a full-scale response. These provocations are not trivial for the junior ally: they impose military readiness burdens, disrupt the economy, generate domestic political pressure, and carry the risk of accidental escalation. Wars typically emerge from sequences of mismanaged crises rather than sudden jumps from peace to con-

flict, so allowing low-level provocations to accumulate unchecked is itself a security concern. The strategic problem facing the junior ally, then, is not whether it will be invaded but how frequently it must absorb dangerous provocations and crises just short of war.

This problem does not look the same from the senior ally's perspective. The senior ally's deterrent posture keeps full-scale war off the table, but it cannot fine-tune the daily risk environment. The senior ally usually maintains a broad strategic posture and faces competing global commitments; it neither can nor does respond to each provocation with tailored policy maneuvers. For the senior ally, any particular adversary's low-level provocations are one regional concern among many. For the junior ally, they are front-page, high-salience events that demand an immediate response. This asymmetry of salience means that the senior ally supplies the strategic deterrent backdrop—the credible threat of catastrophic punishment—while the junior ally is well positioned to shape the adversary's routine behavior below the war threshold. This does not mean the two allies split the sub-war space between them: the senior ally still acts below the threshold, as the analysis will show. The point is that in this space the junior ally is no longer sidelined, and its own policy choices can independently affect the adversary.

The junior ally is structurally well positioned to manage the adversary below the war threshold. Geographic proximity[2] places the junior ally in continuous contact with the adversary and allows faster reactions than the distant patron; the adversary knows this. Proximity also opens a wider window for frequent interaction across military, diplomatic, and economic channels, generating a steady stream of observable policy moves that the adversary monitors. The distant senior ally lacks this continuous presence. In addition, this continuous presence allows the junior ally to accumulate more information about the adversary's intentions and behavior than the senior ally can from afar.

Taken together, the junior ally possesses both the structural capacity (proximity, frequent contact, informational advantage) and the strategic motivation (disproportionate costs of

[2]Examples include Eastern European NATO members facing Russia, the Philippines sharing open sea with China, and South Korea sharing a border with North Korea.

provocations, higher issue salience) to independently manage adversary behavior below the war threshold. It is not primarily trying to deter war; that function is largely accomplished by the patron's strategic posture and extended deterrent threats. It is trying to reduce the frequency and intensity of sub-war provocations that the senior ally, with its broader commitments and greater distance, is less inclined and less able to manage day to day.

> ***H2 (Junior-ally independence).*** *In asymmetric alliances where all-out war is structurally deterred, more cooperative junior-ally policy is associated with fewer sub-war provocations, whereas more coercive junior-ally policy is associated with more provocations, independently of the senior ally's policy.*

4 Data and Methods

In this section, I introduce a novel monthly dataset coding U.S. and South Korean policy orientations toward North Korea across five dimensions, describe how I measure the main outcome of interest, and explain the modeling strategy.

I focus on the US–ROK–DPRK triangle because it is a clear example of an asymmetric alliance, which lets me examine how changes in each ally's policy relate to the adversary's behavior. I structure the dataset as a monthly panel at the month–administration level for the U.S. and South Korea. For each month–administration combination, I code the government's policy orientation toward North Korea across five dimensions. The reasons for my case and time-period choice are threefold. First, I study the period from 1991 onward because it marks the post–Cold War era, when the collapse of the Soviet Union changed the broader strategic environment and prompted adjustments in the US–ROK alliance. This era was marked by recurring missile tests and armed incidents but no escalation to full-scale war, the below-threshold setting the theory addresses. Second, the early 1990s are when North Korea's nuclear and missile programs became central to U.S. and South Korean policy toward Pyongyang, making this a natural starting point for analyzing policy orientations and

DPRK reactions. Third, I end the sample in 2011 to isolate the Kim Jong Il era and avoid confounds from the leadership transition to Kim Jong Un, whose tenure reflects different decision-making and a much higher rate of missile and nuclear testing and is better treated as a separate phase.

4.1 Explanatory Variables: A New Five-Dimension Monthly Dataset of Alliance Policy

My main independent variables are monthly policy-orientation measures for each ally. These measures, and the coding scheme that produces them, represent the paper's primary empirical contribution. The dataset advances beyond existing approaches to alliance behavior in two respects.

First, the measures are substantially more granular than existing operationalizations of alliance behavior, which rely on annual or administration-level indicators such as defense spending or troop levels. Monthly coding captures the tempo of below-the-war-threshold interactions between allies and adversaries, which unfolds far faster than annual aggregates can track. Coding each ally's policy independently, rather than treating the alliance as a unitary actor, makes it possible to test whether the junior ally exerts an independent influence on the adversary: the paper's central question. Furthermore, disaggregating policy into five dimensions allows identification of which specific instruments correlate with adversary responses, avoiding the loss of information that occurs when distinct instruments are collapsed into a single index. The five-point scale within each dimension also captures gradations in policy intensity that a binary hawks-versus-doves framing would miss; for instance, the difference between a moderate conciliatory signal and a landmark peace declaration.

Second, the five dimensions correspond to generic policy instruments that any state employs toward adversaries: economic aid, diplomacy, sanctions, rhetoric, and military posture. Because these categories are not specific to the Korean Peninsula, the dimensional structure is portable to other alliance-adversary triads (e.g., U.S.–Taiwan–China), requiring only the

identification of relevant policy chronologies while the categories, coding scale, and decision rules carry over directly. The coding rubrics specify explicit thresholds and decision rules for each dimension, and they rely on observable policy actions rather than subjective assessments of intent, supporting inter-coder reliability testing and straightforward replication.

To construct these monthly measures, I rely primarily on official South Korean and U.S. government sources that provide detailed chronologies of policy actions toward North Korea. For the ROK side, the main sources are the Ministry of National Defense's 국방연표 (Defense Chronology) and the Ministry of Unification's 남북관계연표 (Chronology of Inter-Korean Relations) and 통일백서 (Unification White Paper), which document aid decisions, inter-Korean talks, sanctions measures, official statements, and military posture changes. For the U.S. side, I draw on the same ROK chronologies for externally visible actions, supplemented by English-language chronologies such as those compiled by the Arms Control Association and the National Security Archive at George Washington University, as well as official U.S. government policy and strategy documents for each administration (e.g., National Security Strategies, Defense Department reports, and State Department releases).

Based on these sources, I construct the independent variables by combining two components: a doctrinal baseline that captures each administration's overall intended posture toward North Korea (B_g), and five policy dimensions that capture monthly policy behavior (X_t).

The doctrinal baseline (B_g) represents each administration's stated strategic orientation toward North Korea, coded from the official sources described above. The baseline takes a value of $+1$ when the administration prioritizes engagement and rapprochement, 0 when the doctrine is balanced or ambiguous, and -1 when it emphasizes confrontation and hostility. For example, Kim Dae-jung's "Sunshine Policy" is coded as $+1$, since it was a clearly cooperative era, while Obama's "Strategic Patience" is coded as 0, reflecting a posture that neither prioritized engagement nor confrontation. This baseline is fixed throughout each administration's tenure, capturing doctrinal intent rather than month-to-month behavior. I

restrict it to the narrow $[-1, +1]$ range to reflect that it represents an underlying orientation, not an active policy implementation.

However, the doctrinal baseline alone cannot capture the significant variation in policy behavior within a single administration. Even under Kim Dae-jung's strongly engagement-oriented presidency, South Korean policy toward North Korea fluctuated between cooperative and hostile signals depending on events; for instance, in July 1999, Kim urged U.S. Defense Secretary William Cohen to support missiles with ranges over 500 km as a deterrent against North Korea's missile development, a move that also sent a confrontational signal toward Pyongyang. To capture this variation, I code five monthly policy dimensions—economic aid (E), diplomacy (D), sanctions (S), rhetoric (R), and military posture (M)—each ranging from -2 (strongly confrontational) to $+2$ (strongly engagement-oriented). Together, these form the monthly behavioral component:

$$X_t = \frac{1}{5}\Big(E_t + D_t + S_t + R_t + M_t\Big) \tag{1}$$

The overall monthly orientation is then computed as $O_t = B_g + X_t$, combining doctrinal intent with observed behavior in a given month. This formulation allows reinforcement and offsetting effects: an engagement-leaning doctrine combined with confrontational actions may yield a neutral net orientation.

As a general coding rule, policies are coded in the month of implementation, and their effects are assumed to decay in subsequent months unless the action is ongoing or renewed. Each dimension ranges from -2 to $+2$, except for economic aid, which takes only non-negative values since restrictive economic measures are captured under sanctions. Although economic aid and sanctions might appear to be a single axis, I maintain their separation because providing new aid and lifting existing sanctions carry different political and communicative implications, even though both indicate cooperative intent. When multiple events occur within a dimension during the same month, I record the strongest one (highest abso-

lute value). Below, I describe each dimension and illustrate the coding with examples from the dataset.

Economic aid captures officially announced or implemented economic assistance or humanitarian aid, coded on the basis of publicly confirmed commitments or disbursements rather than proposals. Scores range from +1 for minor assistance (below $50 million USD or 50 billion KRW) to +2 for major packages (at or above those thresholds).

Diplomacy captures official talks, negotiations, working-level meetings, or summits between the U.S. or South Korea and North Korea, including multilateral formats such as the Six-Party Talks. Higher-level engagement receives higher scores: former President Carter's 1994 visit to Pyongyang scores +2, while routine working-level meetings score +1. Cancellation or suspension of talks is coded negatively: withdrawal from planned negotiations scores −1, while formal termination of dialogue channels would score −2, though no such case appears in the dataset.

Sanctions captures new unilateral restrictive measures, or official easing and relaxation, by the U.S. or South Korea; UN-initiated sanctions are excluded. This dimension measures the direction of policy change in a given month, not the standing level of restrictions already in place. Comprehensive sanctions packages score −2, while narrower or targeted measures score −1. Partial relaxation or temporary exemptions score +1, and substantial lifting or restoration of trade channels scores +2.

Rhetoric captures the tone and content of official government communications, speeches, statements, and designations concerning North Korea. For example, President Roh Tae-woo's July 1991 directive in Vancouver for breakthrough inter-Korean civilian exchanges, issued as part of his Nordpolitik, is coded as +2, a landmark conciliatory statement. Conversely, South Korea's designation of North Korea as the "principal enemy" in its December 2010 Defense White Paper is coded as −2.

Military posture captures joint exercises, military deployments, or readiness changes explicitly tied to North Korea, excluding unrelated global operations. Major new deployments

or large-scale joint exercises score −2, while moderate expansions of exercise scale score −1. Partial postponement or downsizing of exercises scores +1, and complete suspension of joint exercises or troop withdrawals for conciliatory purposes scores +2.

Finally, in addition to each ally's individual measures, I construct an alignment variable to test whether policy coordination between the two allies affects adversary behavior independently of each ally's own policy stance. Larger values indicate greater divergence between allies on that dimension. If the conventional view is correct that the alliance deters as a unit, alignment should matter; if instead each ally operates through its own channel, alignment should add little explanatory power. For each dimension, I calculate the absolute difference between the U.S. and South Korean scores. Perfect coordination yields 0; for example, if U.S. rhetoric is −1 and South Korean rhetoric is +2 in a given month, the alignment score for that dimension is 3.

4.2 Outcome Variable: North Korean Provocations

The theory developed in Section 3 concerns adversary behavior below the war threshold: hostile actions that impose real costs on the alliance in the form of readiness burdens, economic disruption, escalation risk, and domestic political pressure. I operationalize this concept as the monthly count of North Korean provocations. I define provocation here as a discrete, publicly observable act by the DPRK that signals hostile intent toward the U.S., South Korea, or both, while remaining below the level that would be expected to trigger full-scale military retaliation. This definition encompasses a range of actions, including missile and nuclear tests, exchanges of fire, armed infiltrations, naval incursions, bombings, and similar hostile acts, unified by their shared strategic function: probing alliance redlines and imposing costs without crossing the war threshold.

The data are drawn from the CSIS Beyond Parallel provocation dataset, which records North Korean provocations from 1958 to the present using a consistent event-identification protocol based on official government reports, military announcements, and verified news

sources (CSIS Beyond Parallel 2025). The database covers physically implemented hostile acts such as missile and nuclear tests, exchanges of fire, infiltrations, and incursions, and does not code purely rhetorical threats or statements as provocations. I aggregate these individual events into a monthly count for the 1991–2011 sample period.

A count specification rather than a binary provocation/non-provocation indicator is appropriate here because the theoretical question is not simply whether North Korea provokes in a given month but how much adversary activity the alliance faces. If cooperative policy reduces the intensity of sub-war pressure, that effect should appear as a reduction in the number of provocations per month, a distinction that a binary measure would obscure.

Figure 1 displays the monthly distribution of DPRK provocations over the sample period. Two features of the distribution are immediately apparent. First, the data are dominated by zeros: of 252 monthly observations, 194 (77 percent) register no provocations at all, while the remaining months are concentrated at low counts with occasional spikes. The mean monthly count is 0.33 and the variance is 0.60, confirming substantial overdispersion relative to a Poisson process. This structure motivates the negative binomial specification used in the estimation. Second, provocations tend to cluster in time, appearing in short bursts separated by long quiet periods rather than arriving at a steady rate. This pattern is consistent with an adversary that responds to shifting external conditions, in line with the assumption that Pyongyang monitors allied policy signals.

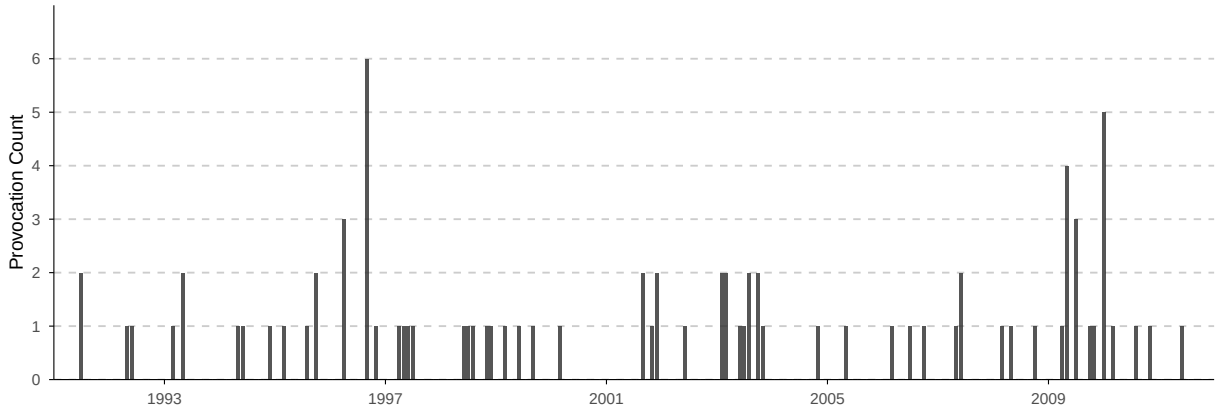


Figure 1: Monthly DPRK Provocation Count, 1991–2011. Data from CSIS Beyond Parallel. Alt text: Bar chart showing monthly counts of North Korean provocations from 1991 to 2011, with most months recording zero and occasional spikes reaching up to six.

4.3 Estimation Strategy

I estimate the relationship between allies' policy orientations and DPRK provocations using negative binomial regression models. As the distribution described above indicates, the dependent variable is overdispersed with a large mass at zero, which violates the equidispersion assumption of the Poisson model and makes the negative binomial specification more appropriate. I use the standard negative binomial rather than a zero-inflated model because the excess zeros in the data have a substantive interpretation: most months produce no provocations precisely because the adversary chose not to act, not because provocations were structurally impossible in those months.

I include all policy-orientation variables with a one-month lag, so that U.S. and ROK policy choices in month $t-1$ are used to predict DPRK provocation counts in month t. The lag reflects the theoretical assumption that policy signals require time to be observed and processed by the adversary before influencing behavior, and it helps mitigate simultaneity concerns. The robustness checks in Section 5 confirm that effects are concentrated at the one-month lag and disappear at two and three months, consistent with Pyongyang responding to recent signals rather than accumulated policy history. The descriptive statistics above

refer to the full 1991–2011 monthly series (252 months); the lagged models drop the first observation, yielding 251 months in the estimation sample.

I estimate separate models using overall policy orientation scores, disaggregated policy dimensions, and alignment measures. First, I estimate bivariate models using each ally's overall policy orientation ($O_t = B_g + X_t$), which combines the administration-level doctrinal baseline with the monthly behavioral component:

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1\, O_{t-1}^{\text{ROK}} \tag{2}$$

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1\, O_{t-1}^{\text{US}} \tag{3}$$

$$\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 + \beta_1\, \text{Alignment}_{t-1} \tag{4}$$

The disaggregated model enters all fifteen dimension-level regressors simultaneously, where each dimension enters as a separate regressor with its own coefficient:

$$\begin{aligned}\log\Big(\mathbb{E}[\text{Provocations}_t]\Big) = \beta_0 \\ &+ \beta_1\,\text{ROK}_{E,t-1} + \beta_2\,\text{ROK}_{D,t-1} + \beta_3\,\text{ROK}_{S,t-1} \\ &+ \beta_4\,\text{ROK}_{R,t-1} + \beta_5\,\text{ROK}_{M,t-1} \\ &+ \beta_6\,\text{US}_{E,t-1} + \beta_7\,\text{US}_{D,t-1} + \beta_8\,\text{US}_{S,t-1} \\ &+ \beta_9\,\text{US}_{R,t-1} + \beta_{10}\,\text{US}_{M,t-1} \\ &+ \beta_{11}\,\text{Align}_{E,t-1} + \beta_{12}\,\text{Align}_{D,t-1} + \beta_{13}\,\text{Align}_{S,t-1} \\ &+ \beta_{14}\,\text{Align}_{R,t-1} + \beta_{15}\,\text{Align}_{M,t-1}\end{aligned} \tag{5}$$

In additional robustness checks reported in the Results section, I examine alternative lag structures and specifications, including tests of co-movement, linearity, and reverse causality tests. The disaggregated model enters only the fifteen dimension-level regressors without

additional controls; I discuss this design choice and potential confounds in Section 5.4.

Figure 2 shows the monthly variation in US and ROK policy across the aggregate orientation and all five dimensions over the full 1991–2011 sample period. The plots demonstrate considerable fluctuation within each ally over time and illustrate the divergence between the two allies, suggesting that the junior ally maintained an independent policy orientation distinct from its patron's. It is also noticeable that sanctions were very rarely used: South Korea recorded only two sanctions-related policy actions toward North Korea over the entire twenty-year period (Ministry of Foreign Affairs 2026).

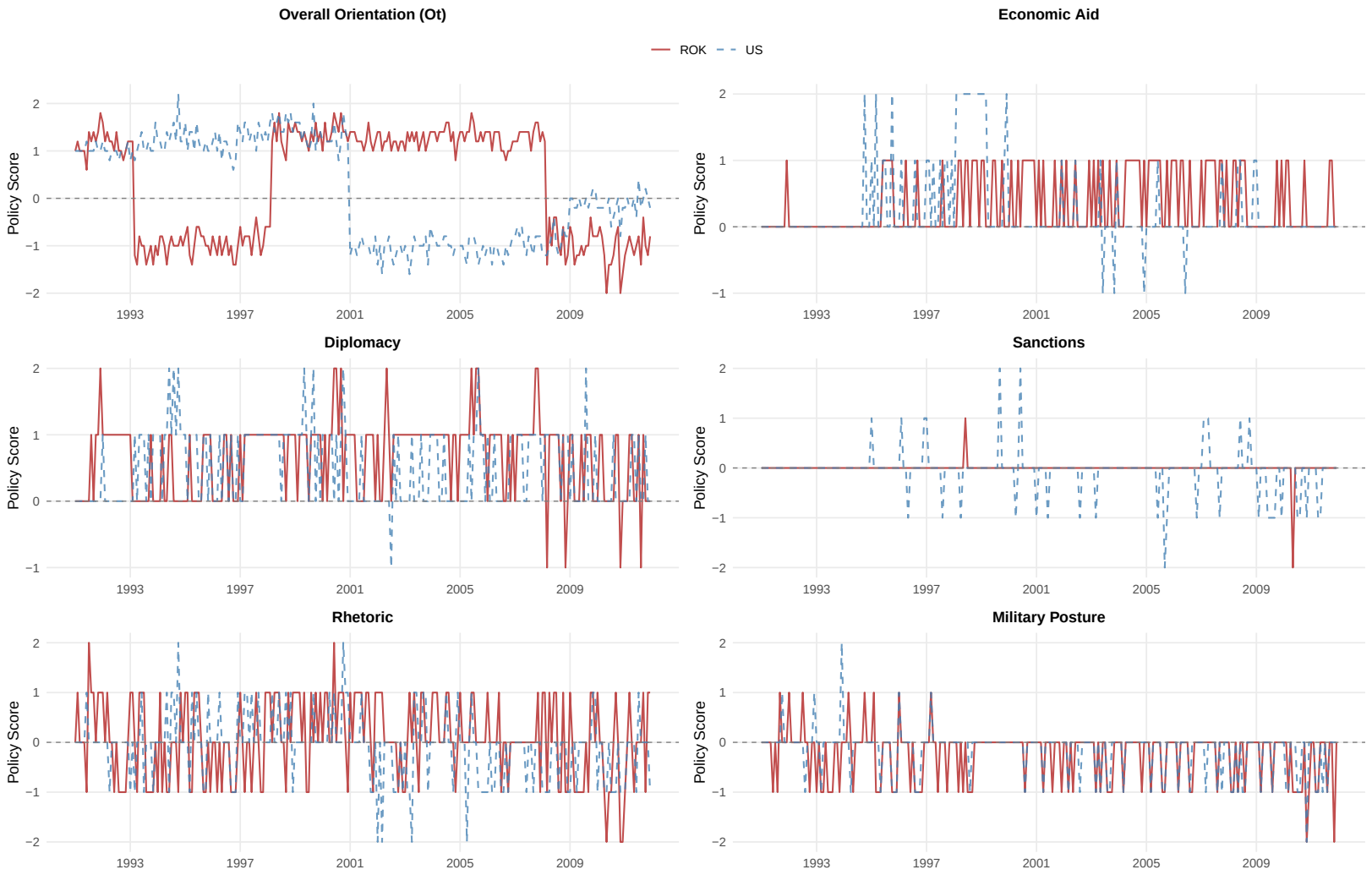


Figure 2: US and ROK Policy Toward North Korea by Dimension, 1991–2011. Positive values indicate engagement-oriented policy; negative values indicate confrontation-oriented policy. Alt text: Six line plots showing monthly US and ROK policy scores toward North Korea from 1991 to 2011 across overall orientation, economic aid, diplomacy, sanctions, rhetoric, and military posture. ROK scores are shown in solid red and US scores in dashed blue.

5 Results and Discussion

5.1 Cooperative policy

H1 expects that more cooperative policy is associated with fewer provocations, whereas more coercive policy is associated with more. The aggregate orientation models offer a first test. When each ally's overall policy orientation (O_t) is entered as a single predictor, the ROK coefficient approaches conventional significance ($p = 0.092$) in the negative direction, indicating that a more cooperative overall posture is associated with fewer provocations (Table 1).

Table 1: Overall Orientation and DPRK Provocations

	ROK	US	Alignment
Overall Orientation (O_t)	$-0.200^{\dagger}$	-0.022	0.123
	(0.119)	(0.130)	(0.160)
Constant	-1.056^{***}	-1.091^{***}	-1.311^{***}
	(0.140)	(0.141)	(0.318)
N	251	251	251

† p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Entries are log-coefficients from negative binomial models. Standard errors in parentheses. Exponentiate coefficients to obtain incidence rate ratios. All independent variables are lagged one month. Dependent variable: monthly DPRK provocation count. Sample: 1991-2011. $O_t = B_g + \bar{X}_t$.

However, because O_t collapses five distinct instruments into a single score, it cannot isolate which specific policy channels are driving this pattern. Disaggregating by dimension clarifies the picture. In the multivariate model with all fifteen dimension-level regressors entered simultaneously (Table 2), a one-unit increase in South Korea's cooperative rhetoric is associated with approximately a 31% reduction in expected DPRK provocations, while a one-unit increase in U.S. diplomatic engagement is associated with approximately a 55% reduction (Figure 3).

Table 2: All Policy Dimensions and DPRK Provocations

		Estimate
ROK	Economic Aid	0.043 (0.379)
	Diplomacy	−0.068 (0.259)
	Sanctions	1.660 (1.544)
	Rhetoric	−0.364* (0.182)
	Military Posture	−0.143 (0.385)
US	Economic Aid	0.081 (0.268)
	Diplomacy	−0.797** (0.287)
	Sanctions	−0.651 (0.549)
	Rhetoric	0.320 (0.222)
	Military Posture	0.071 (0.437)
Align	Economic Aid	0.269 (0.336)
	Diplomacy	−0.135 (0.262)
	Sanctions	−0.267 (0.558)
	Rhetoric	0.111 (0.215)
	Military Posture	−0.199 (0.415)
	Constant	−0.898** (0.326)
	N	251

† p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Entries are log-coefficients (standard errors in parentheses) from negative binomial models. Exponentiate coefficients to obtain incidence rate ratios. All independent variables are lagged one month. Dependent variable: monthly DPRK provocation count. Sample: 1991–2011.

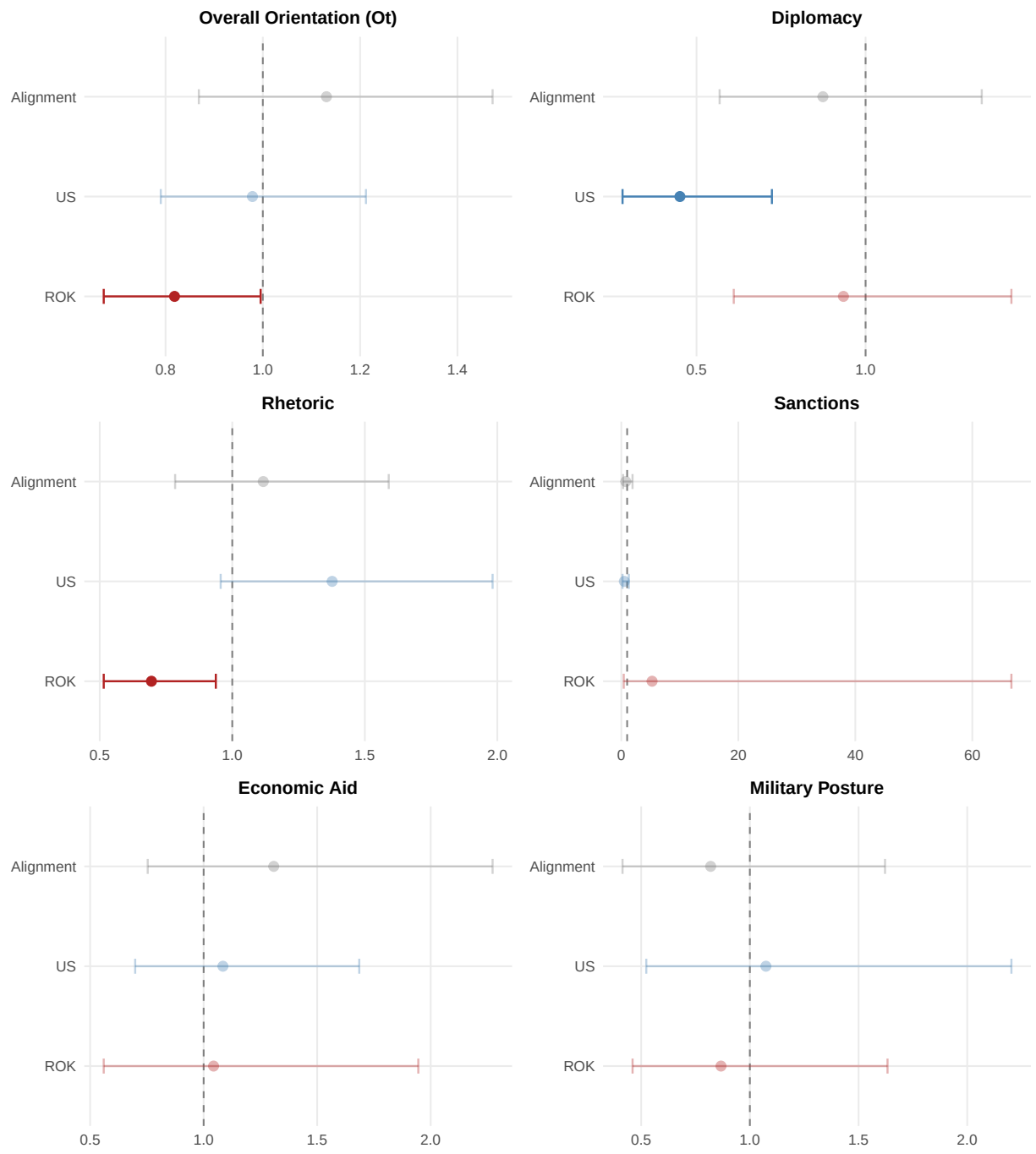


Figure 3: Policy Dimensions and DPRK Provocations. Incidence rate ratios with 90% confidence intervals (IRR < 1 = fewer provocations; IRR > 1 = more provocations). Alt text: Dot plot showing incidence rate ratios with 90% confidence intervals for ROK, US, and alignment across six policy dimensions. Values below one indicate fewer provocations. ROK rhetoric and US diplomacy show statistically significant negative effects.

The remaining dimensions, economic aid, sanctions, and military posture, show no sig-

nificant effects for either ally. The two significant channels survived multiple robustness checks, including tests for co-movement, reverse causality, and alternative lag structures. Both were lagged one month and were not predicted by prior provocations in the reverse causality tests, strongly suggesting that cooperative South Korean rhetoric and US diplomacy preceded, rather than responded to, lower provocation rates from North Korea. This pattern supports H1: under a credible strategic deterrent backdrop, cooperative policy on specific channels that can open an exit for the adversary is associated with fewer provocations, whereas the aggregate orientation measure is not. It is consistent with the assurance literature's emphasis that deterrence depends not only on threats of punishment but on assurances that restraint will be acknowledged (Stein 1985; Kydd and McManus 2017; Pauly 2024). The diplomacy result also converges with CSIS Beyond Parallel's finding of an inverse correlation between US–DPRK negotiations and North Korean provocations (Collins 2017). Why South Korean rhetoric and US diplomacy in particular, rather than the other instruments, is a question I take up in Section 5.3.

The null findings on military posture and sanctions warrant a cautious reading. The models estimate each instrument's average association with provocations, so a null coefficient indicates only that these dimensions show no measurable association with provocations in this sample, in either direction. This is not evidence that these instruments are inert, and the caution cuts both ways: the significant results establish an association along the cooperative-coercive continuum for two channels, not a general effect of coercion or cooperation across all five. This caution is especially warranted for sanctions, which South Korea deployed only twice over the twenty-year period, leaving the estimate imprecise. With that qualification, the insignificance of military posture is consistent with CSIS Beyond Parallel's independent finding that military exercises have no discernible effect on DPRK provocation patterns (Cha et al. 2016), as well as with more recent work showing that joint military exercises fail to deter and may even provoke North Korean hostile responses (Bernhardt and Sukin 2021). The CSIS study also found that the state of US–DPRK diplomatic relations before

exercises was a better predictor of North Korean behavior than the exercises themselves, further reinforcing the primacy of diplomatic context over military signals.

5.2 Junior ally independence

H2 expects that more cooperative junior-ally policy is associated with fewer provocations, independently of the senior ally's policy. The aggregate results already point in this direction: of the three overall orientation models, only South Korea's coefficient approaches significance, while the U.S. and alignment coefficients are substantively small and statistically null (Table 1). The disaggregated model sharpens this finding. South Korea's cooperative rhetoric remains statistically significant in a specification that simultaneously includes all ROK, US, and alignment dimensions (Table 2). That is, even after controlling for the full range of US policy instruments—economic aid, diplomacy, sanctions, rhetoric, and military posture—South Korea's rhetoric toward North Korea retains its independent association with lower provocation rates. This finding challenges the conventional wisdom that junior allies are passive recipients of security rather than active contributors to adversary management.

The mechanisms through which the junior ally can independently influence the adversary's strategic behavior are geographic proximity and the higher salience of adversary management for the junior ally compared to its senior partner. South Korea shares a land border with North Korea while the US is geographically distant, and managing Pyongyang is a top-tier security priority for Seoul in a way it is not for Washington, which faces competing global commitments. Pyongyang understands this asymmetry: North Korea recognizes that South Korea is directly affected by its strategic moves and that Seoul actively pursues its own policy toward Pyongyang rather than simply acting as a US proxy.

The 2001–2008 period illustrates this dynamic concretely. When the Bush administration adopted a confrontation-leaning posture toward Pyongyang, marked by the January 2002 "Axis of Evil" designation alongside sustained hostile rhetoric, expanded sanctions, and the suspension of the Agreed Framework, South Korea's Kim Dae-jung and Roh Moo-

hyun administrations moved in the opposite direction. Seoul continued to frame North Korea as a partner for engagement, emphasizing dialogue, and inter-Korean cooperation, even as Washington labeled DPRK a threat. This produced a sustained period of policy divergence visible in Figure 2. During these years, DPRK provocations remained relatively low despite the sharp deterioration in US–DPRK relations, consistent with the statistical finding that it is the junior ally's rhetorical posture, not the senior ally's coercive stance, that tracks provocation behavior. Had the senior ally's confrontational turn been the dominant signal, we would expect provocations to have increased as US policy hardened; instead, the pattern is more consistent with Pyongyang responding to the proximate ally's cooperative signals. This does not establish a monocausal link; North Korea's behavior during this period reflected multiple considerations. Nevertheless, the co-movement between Seoul's rhetorical orientation and Pyongyang's provocation rate, even as Washington moved in the opposite direction, is difficult to reconcile with a purely patron-centric account of adversary behavior.

The null results from alliance alignment reinforce this point from a different angle (Table 1 and Table 2). The fact that alignment is null while ROK rhetoric alone is significant suggests that it is the junior ally's own policy stance, not its coordination with the senior ally, that shapes adversary behavior. Not only does ROK policy survive controlling for US policy, but the degree of policy divergence between the two allies adds nothing to the model.

5.3 Division of labor

The preceding results suggest that cooperative policy is associated with fewer provocations and that the junior ally contributes independently, but they raise a further question: why rhetoric from South Korea and diplomacy from the US, not any other tools? I interpret this pattern as a division of labor between the senior and junior allies. Diplomacy is a higher-profile, resource-intensive tool (summits, envoys, formal negotiations) while rhetoric is a more granular, everyday signal; and this maps onto the structural difference between what the senior ally and junior ally each bring to the table. The pattern is consistent

with a division of labor in which U.S. diplomacy helps set the strategic framework, while South Korean rhetoric helps manage the day-to-day political temperature. One way to read this pattern is through the three conditions for effective assurance set out in Section 3.2, though this interpretation is exploratory rather than a direct test. ROK rhetoric and US diplomacy plausibly satisfy all three. Both are observable: they are public, frequent, and legible to Pyongyang as deliberate policy. Both reflect genuine discretion: Seoul controls its own rhetorical posture toward the North, and Washington controls whether it engages Pyongyang's core security concerns through formal talks. And both map restraint onto a visible reciprocal pathway: rhetoric signals whether the immediate environment is hospitable, while diplomacy signals a route toward the recognition and security guarantees North Korea seeks. I do not claim that the remaining instruments fail these conditions; the null estimates cannot establish that, particularly for sanctions, which are rarely observed in the sample. The criteria are offered to make sense of why the two significant channels might carry assuring content, not to convict the others.

Specifically, US diplomacy matters because it is the one US instrument that directly addresses what North Korea cares about most from Washington: security guarantees and regime recognition (Byman and Lind 2010; Howell 2020). Formal talks and summits signal that the US is willing to engage on Pyongyang's existential concerns. North Korea's primary grievance with the US is strategic and security-oriented, so diplomatic engagement is the channel through which Washington can credibly signal that restraint will be rewarded. ROK rhetoric, on the other hand, matters because South Korea is the geographically proximate, directly affected party whose day-to-day statements Pyongyang monitors as an indicator of the immediate security environment. ROK rhetoric is low-cost, high-frequency, and adjustable month-to-month, making it well suited to managing sub-war dynamics. Rhetoric is often characterized as "cheap talk" because it is costless to produce and non-binding. Yet as Sartori (2002) argues, diplomatic communication carries reputational costs that give it credibility beyond its material price. The finding that ROK rhetoric is associated with

fewer provocations is consistent with this logic: when the adversary is already monitoring the proximate ally for any indication of the immediate security environment, the content of those signals matters because misrepresentation would damage Seoul's credibility in future interactions with Pyongyang. Pyongyang treats Seoul's rhetorical posture as an indicator of whether the immediate neighborhood is hospitable or hostile to its interests.

Furthermore, the Granger causality and cross-correlation tests showed that these two channels move independently rather than in coordination, which is consistent with the alignment null result. Each ally operates through its own channel, not through joint action. Both ROK rhetoric and US diplomacy are among the most frequently deployed instruments in their respective toolkits, which means they also generate the most consistent stream of signals for Pyongyang to observe.

5.4 Robustness and additional specifications

To address concerns that temporal dependence in provocations may be driving the results, I estimate models that incorporate lagged outcomes and alternative lag structures. Two findings emerge. First, adding the prior month's provocation count as a control does not change the significance of ROK rhetoric or US diplomacy, and the lagged dependent variable itself is insignificant, suggesting that serial correlation is not driving the results. Second, effects are concentrated at the one-month lag and disappear at two and three months. This pattern is consistent with Pyongyang responding to recent signals rather than accumulated policy history, reinforcing the signaling interpretation. The core findings are also robust to tests for co-movement between the two significant channels, linearity, and multicollinearity; full results appear in the Appendix.

The disaggregated model is deliberately parsimonious, entering only the fifteen dimension-level regressors without additional controls. Many variables that might appear to be omitted—such as diplomatic engagement, sanctions regimes, or military exercises—are already captured within the five-dimension coding scheme, which was designed to be comprehensive.

The doctrinal baseline (B_g) absorbs administration-level variation on the allied side through the aggregate orientation models, and the sample restriction to 1991–2011 effectively holds constant the DPRK leadership era by excluding the structurally different Kim Jong-un period.

To address potential confounds external to the coding scheme, I test a binary indicator for months containing major DPRK national holidays and a linear time trend; both are insignificant in bivariate specifications, and adding them to the full model individually or in combination does not change the significance of ROK rhetoric or US diplomacy (Appendix). China's influence is partially captured through its downstream effects: when Beijing facilitates or obstructs multilateral diplomacy, e.g., the Six-Party Talks, the observable consequences register in the US and ROK diplomacy dimensions. Taken together, the comprehensive coding scheme, the doctrinal baseline, the sample restriction, and the holiday and time-trend tests address the most important confounds. While these checks do not eliminate the possibility of omitted variable bias, they suggest that the results are not straightforwardly attributable to an unobserved third factor.

6 Conclusion

This paper has asked how allies' policy orientations toward the adversary shape provocations in asymmetric alliances once all-out war is already off the table, and whether junior allies exert an influence of their own. Focusing on the U.S.–ROK alliance and North Korean provocations from 1991 to 2011, it argued that in this below-the-war-threshold space, assurance-bearing cooperative policy is associated with fewer provocations while added coercion is less likely to suppress them, and that the junior ally's policy can shape this likelihood independently of the senior ally. Using an original monthly dataset that disaggregates U.S. and South Korean policies across five instruments, the analysis showed that at the aggregate level, only South Korea's overall orientation approaches significance, while U.S. policy and

inter-ally alignment show no effects. Disaggregating by dimension, South Korea's cooperative rhetoric and U.S. diplomatic engagement are the only channels consistently associated with fewer North Korean provocations. South Korea's rhetorical stance remains significant even after controlling for the full range of U.S. policy instruments, indicating that a junior ally can independently influence adversary behavior through its own cooperative policies rather than merely transmitting the patron's posture.

These findings speak directly to the gaps identified in Section 2 and carry two main implications for scholarship on alliances and extended deterrence. First, they show that assurance, not only threat, is central to managing an adversary below the war threshold: once coercive pressure is saturated, what shapes provocations is cooperative policy that makes restraint worthwhile for the adversary. Second, they show that assurance need not come from the patron. Because the junior ally is structurally suited to provide it, disaggregating both allies and their policy instruments reveals channels of assurance that treating alliances as unitary actors, or focusing on the senior ally's capabilities and resolve, would obscure. In this sub-war setting, deterrence depends not only on the patron's ability to threaten punishment but on how each ally assures the adversary that cooperation will be returned.

The analysis also points to concrete policy implications for alliance management and burden sharing. Current burden-sharing debates typically center on aggregate inputs such as defense spending levels or troop contributions, implicitly treating more capability as the primary way junior allies contribute to deterrence. The evidence here suggests a different division of labor may be more efficient: senior allies can continue to provide the bulk of coercive capability and high-profile diplomacy, while junior allies play a deliberate role in day-to-day cooperative engagement, especially through rhetorical signaling. Rather than pushing junior allies only to spend more or match the patron's coercive posture, policymakers could instead encourage and empower them to use engagement instruments that they are structurally well positioned to wield, thereby improving deterrence performance below the war threshold without simply scaling up aggregate expenditure.

These findings should be interpreted with appropriate caution. The analysis focuses on a single asymmetric alliance and one adversary, which limits the ability to generalize without further cross-case testing. The policy measures are coded from historical records by a single researcher, which, despite a transparent coding scheme, raises the possibility of measurement error that future work could address through replication or multi-coder designs. All provocations are also weighted equally, though future work could differentiate between types of actions, for example assigning greater weight to nuclear tests than to minor incursions. In addition, the sample ends in 2011 and therefore does not capture the Kim Jong Un era or the more recent Moon–Yoon and Biden-period dynamics that motivated the opening puzzle. Preliminary analysis extending the coding through 2024 is reassuring on this point: cooperative South Korean rhetoric and U.S. diplomatic engagement remain associated with fewer provocations in the extended sample, while a large and significant Kim Jong Un–era indicator confirms that the period differs structurally enough to warrant the sample restriction adopted here and separate theoretical treatment in future work. These constraints mean the paper cannot claim that junior-ally cooperative policy will always reduce provocations across alliances and time, but they do show that under the conditions studied, junior allies can play a meaningful and previously overlooked role.

The framework and measurement strategy developed here open several avenues for future research. The coding scheme itself is designed to be portable: because it relies on observable policy events along five standardized dimensions rather than case-specific knowledge, it can be applied to any asymmetric alliance with a shared adversary. A natural next step is to extend the analysis to the U.S.–Taiwan relationship. Although Taiwan lacks a formal alliance with the U.S. and its statehood remains contested, the structural conditions that drive the theory are present: geographic proximity to the adversary, high issue salience, frequent cross-strait interaction, and a de facto U.S. security commitment through the Taiwan Relations Act. Taiwan's independent policy swings between engagement and confrontation toward Beijing, China's sustained sub-war provocations, and a high-cost escalation environment closely

mirror the conditions on the Korean Peninsula; applying disaggregated monthly coding to Taiwanese and U.S. policies toward China would allow a direct test of whether the framework travels beyond formal alliances to de facto security partnerships (Bush 2005; Stokes 2023). A more theory-stretching application would examine Turkey's role within NATO vis-à-vis Russia, where Ankara combines alliance protection with substantial independent engagement of Moscow, raising questions about how the framework should treat cases in which a junior ally's policy toward the adversary mixes risk management with economic self-interest and strategic autonomy (Kutlay and Öniş 2021; Kara 2024).

Across these and other extensions, the core message remains the same: to understand how adversaries are managed below the war threshold, we must look not only at what powerful patrons do, but also at how junior allies use their own cooperative policies to shape adversary behavior day to day. Recognizing that role helps move debates about alliance politics and extended deterrence beyond aggregate capabilities and toward a more precise understanding of how specific allies, through specific policies, shape adversary behavior below the war threshold.

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